Research Question	Approach / Steps to Answer the Question	Tools & Libraries Used
RQ1: Can we predict heart disease risk using machine learning?	Clean and preprocess data; split into train/test sets; train Logistic Regression, Random Forest, and XGBoost models; compare performance; select the best model.	Python, Jupyter Notebook, pandas, numpy, scikit-learn, XGBoost, matplotlib, seaborn
RQ2: Which factors are most important for predicting heart disease?	Perform correlation analysis; train tree-based models; extract feature importance; use SHAP values; rank features	pandas, numpy, scikit-learn, XGBoost, SHAP, seaborn, matplotlib
RQ3: Does smoking increase predicted heart disease risk?	Separate smokers and non-smokers; compare disease prevalence; visualize distributions; examine smoking-related variables.	pandas, seaborn, matplotlib, scipy, Sweetviz
RQ4: How does blood pressure affect heart disease risk?	Analyze systolic and diastolic BP distributions; calculate correlations; visualize BP-risk relationships; use partial dependence plots.	pandas, seaborn, matplotlib, scikit-learn
RQ5: Does cholesterol improve prediction accuracy?	Build a baseline model; remove cholesterol-related features; retrain models; compare performance.	pandas, numpy, scikit-learn
RQ6: What role do glucose levels and diabetes play?	Group patients by diabetes status; compare glucose distributions; analyze disease prevalence; assess model predictions.	pandas, seaborn, matplotlib, scikit-learn

RQ7: Which machine learning model performs best?	Train multiple algorithms; evaluate using common metrics; compare results; identify the most reliable model.	scikit-learn, XGBoost, pandas, numpy
RQ8: Does class imbalance affect predictions?	Analyze class distribution; train the model on the original data; apply SMOTE and class weighting; compare results.	imbalanced-learn (SMOTE), scikit-learn, pandas
RQ9: Can we simplify the model without losing accuracy?	Apply feature selection, remove less important variables, build reduced models, and compare with full models.	scikit-learn (RFE, Lasso), pandas, numpy
RQ10: Does the model work equally well for different patient groups?	Divide patients by age, gender, and risk groups; evaluate model performance separately; compare metrics.	pandas, scikit-learn, seaborn, matplotlib